Global Facilities – General
Plastic Piping Pressure Testing Standard

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ECN History **Submit a Change Request**

1.0 Purpose

1.1 The purpose of this Pressure Testing Standard is to provide Micron employees, contractors and vendors with the requirements and guidelines to follow while performing quality inspection and pressure testing for Plastic Piping Installations.

1.2 The document is intended to supplement the Work Procedures and Specifications applicable to the Engineering, Design, and Construction work at Micron Sites including Basebuild Installations, Tool Hookup Installations, Greenfield Construction Projects, and all other Construction activities.

1.3 A detailed Work Procedure shall be developed for each specific pressure testing activity, and shall include the specific means and methods used to execute the pressure test.

2.0 Scope

2.1 System(s) impacted

2.1.1 This document applies to **Plastic Piping (PVC, CPVC, ABS, PE, HDPE, PP, and PVDF)** used for all Facilities systems including, but not limited to, the following systems: UPW/DI Systems, City Water/Industrial Water/Non Potable Water Systems, Chemical Supply and Waste Drain Systems, PCW Systems, Double Containment Piping Systems, Plumbing Systems, Process Vacuum, etc.

2.2 Site(s) and Person(s) impacted

2.2.1 This document applies to all Micron employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
ABS	Acrylonitrile-Butadiene-Styrene piping material
COHE	Control of Hazardous Energy
CPVC	Chlorinated Polyvinyl Chloride piping material
HDPE	High Density Polyethylene piping material
ITP	Inspection and Test Plan
JHA	Job Hazard Analysis
LOTO	Lockout / Tagout
PE	Polyethylene piping material
PP	Polypropylene piping material
PPE	Personal Protective Equipment
PVC	Polyvinyl Chloride piping material
PVDF	Polyvinylidene Fluoride piping material
QA/QC	Quality Assurance / Quality Control
RA	Risk Assessment
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT	• Develop and maintain Standard • Ensure Best Known Methods (BKM) and lessons learned are documented in Standard • Support the Site teams by providing knowledge and technical expertise • Perform periodic document review and update Standard
Global EHS	• Review, approve, and provide technical feedback on Standard • Support the Site teams by providing knowledge and technical expertise
Site Facilities and Project Construction Teams	• Review and provide technical feedback on Standard • Incorporate Standard into all Engineering, Design, and Construction work at Sites and for Projects • Responsible for understanding and following the Standard • Responsible for oversight and coordination with Construction Contractors and Vendors • Share new lessons learned to GFTT Lead and Global EHS, to determine if it is a BKM that can be captured in future revision of Standard
Construction Contractors and Vendors	• Responsible for understanding and following the Standard • Responsible to perform quality control measures to ensure that installations are in accordance with Codes, Specifications, and Micron Requirements • Responsible for setting up, coordinating, and completing system pressure tests • Responsible for documentation and record keeping

5.0 Safety

5.1 The information below outlines important safety components regarding this procedure.

5.1.1 All piping manufacturer's requirements and recommendations shall be followed, including, but not limited to, the following:

- Material ratings for pressure and temperature
- Procedure for Installation
- Procedure for QA/QC, Inspection, and Examination
- Procedure for Preparations prior to Pressure Testing
- Procedure for Pressure Testing

5.1.2 The following are the minimum Personal Protective Equipment (PPE) requirements, that shall be used during the pressure tests:

- Safety Helmet
- Safety Glasses
- Safety Shoes/Boots
- High-Visibility Safety Vest
- In all cases, follow the site-specific requirements, project-specific requirements, and any specific RA/JHA requirements

5.1.3 An RA/JHA shall be developed for the Pressure Testing activity, and shall be reviewed and approved by the Site EHS or Micron Construction Safety Team Member.

6.0 Pressure Testing

6.1 Acceptable Pressure Testing Methods

6.1.1 Hydrostatic testing with water is an acceptable method of pressure testing of Plastic piping systems. The testing pressure must be maintained within the material's pressure rating.

6.1.2 Vacuum testing at negative pressure is an acceptable method of leak testing of Plastic piping systems, in-lieu of Hydrostatic testing at positive pressure. Or the Vacuum testing can be performed as a pre-test for gross leaks, prior to the Hydrostatic test. The testing pressure must be maintained within the material's pressure rating. **Note**: Vacuum testing may not be as effective at identifying leaks, in comparison to Hydrostatic testing. If using vacuum testing, the contractor is still responsible for identifying, containing, and repairing all leaks.

6.1.3 Plastic piping systems shall **NOT** be pressurized using compressed air or other compressed gases, during pressure testing or during actual operation, **except** for cases in which the following exceptions apply:

Exceptions: Plastic piping systems can be pressure tested using compressed air or other compressed gases, **ONLY IF** there is written documentation from manufacturer allowing this activity, **AND** all of the following requirements are followed:

1. Manufacturer's literature shall be reviewed and confirmed for the material pressure ratings and manufacturer's recommendations. Manufacturer shall have written documentation stating that testing with compressed air or gas is allowed.

2. If the manufacturer explicitly states that pressure testing shall not be done, or is not recommended to do, with compressed air or gases, then this instruction shall be followed. This instruction shall take precedence over all material ratings.

3. Test pressures must always be maintained within the material's pressure ratings and manufacturer's recommendations.

4. Proper safety precautions and engineering controls shall be provided, including, but not limited to, the following:
    - Pressure regulator shall be provided, to reduce the supply pressure to within the test requirements
    - Calibrated Pressure relief valve shall be provided, to prevent over-pressurization

5. An RA/JHA shall be reviewed and approved by Site EHS or Micron Construction Safety Team Member, with additional review and approval by GFTT and Global EHS.

6. Some examples of Plastic piping systems rated and approved by the manufacturer for testing and/or use with compressed air include:
    - Asahi "Air-Pro" HDPE Piping
    - GF "Contain-It" PVC Double Containment

6.1.4 **NOTE:** Many manufacturers, industry references, and/or local jurisdictions prohibit the testing or operation of Plastic piping systems using compressed air or other compressed gases.

The following example is for PVC and CPVC piping:

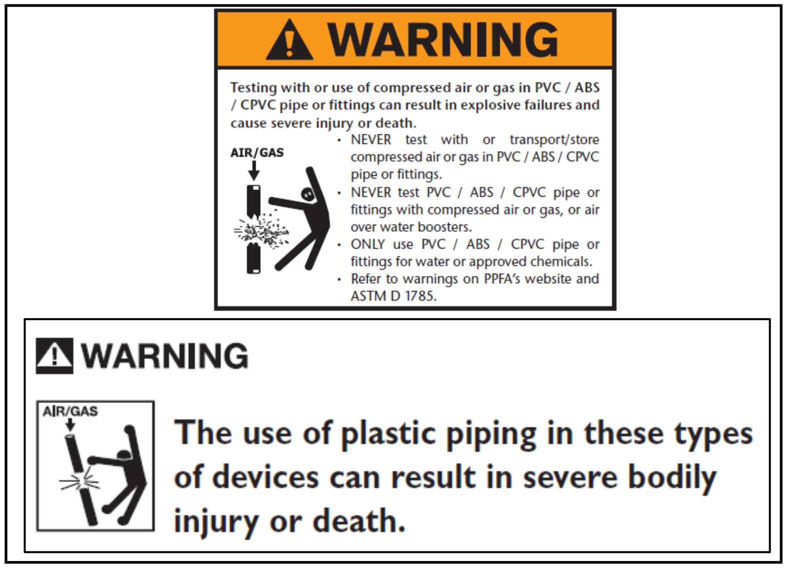
A. For rigid PVC and CPVC piping, most manufacturers have documented that they do not allow the testing or operation of the piping with compressed air or other compressed gases.

B. Certain industry references, including Plastic Pipe Institute (PPI), American Society for Testing and Materials (ASTM), and Occupational Safety and Health Agency (OSHA), do not allow the testing or operation of certain plastic piping (PVC, CPVC, etc.) with compressed air or other compressed gases.

C. The reasons for these restrictions are as follows:

- PVC and CPVC are brittle materials that can be broken or shattered due to applied forces.
- The compressibility of air and gas results in tremendous amounts of stored energy, even at lower pressures.
- A failure of brittle piping material such as PVC or CPVC, combined with the sudden release of stored energy, may result in fragments of broken plastic being thrown at high velocity in multiple directions (i.e. projectiles of shrapnel)
- This scenario creates a substantial hazard to personnel and property within the vicinity of the piping.

Figure 1: Example Manufacturer's Restriction for Testing of Compressed Air for PVC/CVPC/ABS



The following example is for PP and PVDF piping:

A. For rigid PP and PVDF piping, most manufacturers have documented that they do not allow the testing or operation of the piping with compressed air or other compressed gases.

Figure 2: Example Manufacturer's Restriction for Testing of Compressed Air for PP/PVDF

Nibco/Chemtrol PP/PVDF Piping:

WARNING: DO NOT USE OR TEST THE PRODUCTS IN THIS CATALOG WITH COMPRESSED AIR OR OTHER GASES. FAILURE TO FOLLOW THIS WARNING CAN RESULT IN PERSONAL INJURY OR DAMAGE TO PROPERTY.

Georg Fischer (GF) PP/PVDF Piping:

Testing with gases is not recommended.

Asahi PP/PVDF Piping:

For compressed air piping, Asahi/America recommends only the use of manufacturer approved materials for this application. In particular, the Air-Pro system from Asahi is the only system recommended for use in this application due to safety concerns. Air-Pro has been specifically designed and tested for the application of compressed air and gases. All other materials such as PVC, C-PVC, PVDF, and polypropylene are not recommended for compressed air and are not warranted by Asahi/America for that service.

6.2 General Pressure Testing Requirements

6.2.1 All pressure testing activities shall be scheduled and coordinated with Micron.

6.2.2 All testing shall be performed in such a way that it does not interfere with the existing Site operations or other ongoing work. Equipment and material used for pressure testing shall not come into contact with or impact existing systems or components.

6.2.3 **A detailed test procedure shall be developed for each pressure testing activity.**

The requirements for testing pressure and time duration shall be determined specifically for each pressure test activity. The criteria includes, but is not limited to, the following:

- Type of utility (i.e. UPW, Waste Drains, Plumbing, etc.)
- Design operating pressure and temperature
- Piping material and material ratings
- Project requirements,
- Local codes and regulations

The following considerations shall be included:

- The maximum test pressure shall not to exceed 1.5 times the system design pressure.
- The maximum test pressure shall not exceed the pressure rating of any piping, material, devices, or components being tested. Temporarily remove or isolate any devices or components with lower pressure ratings that the test pressure.

6.2.4 Pressure testing may be conducted on the entire piping system at one time, or on smaller sections of the piping individually.

6.2.5 Stop all work on the piping during the entire duration of the pressure test.

6.2.6 Any proprietary liquid testing media or solution shall be approved by Micron before being used to troubleshoot leak locations.

6.2.7 Contractor shall provide all necessary instruments, gauges, and recorders to monitor the test. Pressure gauges shall be calibrated, with adequate pressure range for the test pressure.

6.2.8 Contractor shall provide all materials, components, and devices required to perform the testing, such as hydronic pumps, hydronic hoses, vacuum pumps, vacuum hoses, blanks, plugs, vents, gaskets, and drain valves. Blank flanges, blank plates, etc. shall be selected to withstand the test pressure.

6.3 Pre-Test Procedure Requirements

6.3.1 Review the design/construction documents, manufacturer's published information, and material pressure ratings to ensure the test pressure will not exceed the pressure rating or over-pressurize any equipment or piping within the pressure test boundary.

6.3.2 **An RA/JHA shall be developed for the specific pressure testing activity, and shall be reviewed and approved by the Site EHS or Micron Construction Safety Team Member.**

Also, a detailed test procedure shall be developed for each pressure testing activity, and shall be reviewed and approved by Micron Site Representative and Project System Owner.

The RA/JHA and detailed test procedure shall include, but not limited to, the following considerations:

- Test pressure and time duration
- Piping material pressure rating
- System design operating pressure
- PPE to be used
- Safety precautions and engineered controls (relief valves, etc.) to prevent overpressure
- Area barricades and signage to prevent other personnel from entering the test area
- Other potential hazards including Electrical sources (support equipment, electrical panels, etc.)
- Lockout/Tagout (LOTO/COHE) of any potential sources of hazardous energy
- Lockout/Tagout (LOTO/COHE) of any valves used to isolate the test sections, if required
- Established locations where personnel are witnessing and monitoring the test

6.3.3 Ensure that the installation of all piping is accurate and complete, including, but not limited to, the following checks:

- Review of piping installation versus layout drawings, schematics, and P&IDs
- All joints and flanges are completed and tightened, with no missing bolts or gaskets
- All heat fusion joints and flared joints are completely sealed and cooled
- All piping supports are installed. Piping supports shall be sufficient to support the weight of test medium
- All valve and component orientation and directions are correct
- Proper air vents are installed. Air Vent connections shall be located at high points of piping, or at points where air can be potentially trapped
- Proper drains are installed. Drain connections shall be located at low points of piping, or at points where water can be potentially trapped

6.3.4 Identify all devices and components that will NOT be included in the pressure test. Provide temporary provisions to isolate or remove these devices and components, including, but not limited to, the following:

- Any components that cannot be tested with flow-through (i.e. check valves, regulators, etc.) shall be replaced with a temporary spool piece during the test
- All instrumentation not rated for the test pressure shall be isolated or removed before testing
- Control valves not rated for the test pressure shall be isolated or removed before testing
- Vendor supplied equipment shall not be pressure tested. All such vendor supplied equipment shall have its connections blanked off during the pressure test.
- Relief valves and safety valves shall be blanked off at the inlet flange of the valves. Screwed connection relief and safety valves shall be removed and replaced with plugs or capped ends
- Restrictions to flow, such as orifice plates and mixing nozzles, shall not be installed or shall be removed. Where necessary, items removed should be replaced with temporary spool pieces provided by the contractor

6.3.5 Secure piping and fittings in its position, to prevent excessive movement or damage during testing.

6.3.6 The equipment and source points of connection are the only two acceptable joints to be excluded from a complete system test.

6.3.7 Insulation shall not be installed over any joints in the piping system prior to the test. All flanges, threaded joints and field welds shall be left bare of insulation and free from all visual obstructions.

6.3.8 Perform the following temporary measures for the duration of the test:

- Ensure all equipment, components, or devices not to be tested is disconnected or isolated by closing valves, installing testing blinds, or removing the items
- Ensure all valves inside the test boundary are in the open position
- Ensure Lockout/Tagout (if required) of any closed valves used to isolate the test boundaries
- Ensure isolation valves are closed, for the valves that are used to isolate devices, components, equipment, or instrumentation that is not being tested
- Ensure all future valve points of connection shall be capped or plugged during testing
- Provide temporary restraints for all expansion joints, to withstand the added pressure load. If temporary restraints are not practical, then expansion joints should be removed and either part isolated from the system, or a temporary spool piece for the test installed
- Provide temporary piping supports, if required to sustain the weight of the test media fluid
- Provide temporary piping supports or blocks, for piping that is supported by springs or counter-weights
- If there is any potential risk for freezing, provide the necessary measures to prevent freezing
- Lines containing check valves shall have the source of pressure test applied to the upstream side of the valve. If this is not possible, swing check valves shall be blocked open or their flapper removed. Ball and piston check valves shall have their internal parts removed. Should some or all of these measures be impractical, then check valves shall be blocked off or completely removed

6.3.9 The pressure test equipment and all connections to the test section shall be examined, to ensure that all are in proper operating condition and tightly connected.

6.3.10 The pressure test equipment shall include a pressure relief valve to protect against overpressure of the system, or pressure increases as a result of ambient temperature increases

6.3.11 Perform a physical walkthrough inspection and examination of the entire piping system.

6.4 Test Procedure Requirements

6.4.1 **Per sections 6.2.3 and 6.3.2 above, a detailed test procedure shall be developed for each pressure testing activity.** Follow the detailed test procedure that was developed for the specific pressure testing activity.

6.4.2 **Hydrostatic Pressure Testing –** For Hydrostatic pressure tests, the following precautions shall be considered and/or followed during the test procedure (if applicable or required):

- Piping and equipment (where applicable) shall be thoroughly vented of air before the final Hydrostatic test pressure is applied. Eliminate all trapped air from piping before pressurizing the piping
- To allow air to escape from the test section, flow velocities during filling should not exceed the capacities of air release devices or openings used to release trapped air. To avoid or limit transient pressure surges, the filling flow velocity should not exceed the design velocity of the piping system
- Add make-up water as necessary to achieve the test pressure
- When the test section is completely filled and purged of air, gradually increase pressure in the test section to the required test pressure
- During the water filling and pressurization process, perform visual checks on the test section for leaks
- Allow the test section and the test liquid to equalize to a common temperature

6.4.3 **Vacuum Leak Testing –** For Vacuum leak tests, the following precautions shall be considered and/or followed during the test procedure (if applicable or required):

- The vacuum source used for testing can be provided by Facilities Process Vacuum System/equipment (if available) or portable vacuum pump equipment
- If there is a pressure rise in the test section which indicates a leak, perform visual checks on all joints to identify the leak
- To help identify the leak, an acceptable solution, such as bubble solution or cream, may be applied over the joints. The solution shall be compatible with the piping material, components, equipment, and system. Per section 6.2.6,

any proprietary liquid testing media or solution shall be approved by Micron before being used to troubleshoot leak locations.

6.4.4 **All Testing -** If the test pressure cannot be achieved, or it takes an unreasonably long time to reach the test pressure, there may be faults such as excessive leakage, entrapped air, open valves, or the pressurizing equipment may be inadequate for the size of the test section. If such faults exist, discontinue pressurizing, and correct the faults before continuing the pressure test.

6.4.5 Do not attempt to correct or fix leaks while the test section is under pressure.

6.4.6 If there are faults such as leaks that need repair, de-pressurize the test section and correct the faults or leaks, before re-starting the pressure test.

6.4.7 The test section shall be de-pressurized by reducing pressure or releasing test liquid at a controlled rate. Sudden de-pressurization can cause water hammer.

6.4.8 If no visual leakage is observed, and pressure during the test pressure remains steady for the testing duration, the test is considered accepted.

6.4.9 Report any defects, deficiencies, or abnormal conditions during the test.

6.5 Post-Test Requirements

6.5.1 Upon completion of pressure testing, restore all piping, components, and devices back to their normal configuration and final state:

- Remove all temporary blank flanges, blank plates, and blinds
- Remove all temporary spool pieces and re-install devices and components that were removed for testing (control valves, check valves, etc.). Ensure that the proper flow direction is maintained
- Remove all temporary piping supports and blocks. Inspect the piping for any distortion or abnormalities
- Remove any restraints for the expansion joints

- Valves which were closed for pressure testing and/or draining shall be returned to the proper position
- Un-isolate or re-install all instrumentation, relief valves, and other devices or components
- Re-connect all equipment that was disconnected. Where an equipment has been blanked off at a flanged joint to allow pressure testing of connecting piping, it shall not be necessary to pressure test the joint after the blank is removed. Such joints shall be visually examined to ensure that gaskets are properly installed, bolts correctly tightened, etc.

6.5.2 Install new gaskets for any flanged joints that are temporarily removed or opened-up during the pressure test. Do not re-use any gaskets on opened sections of piping.

6.5.3 Disconnect the test equipment, test pump, test gauges, and test safety valves, and remove these items.

6.5.4 Provide a pressure test report documenting all sign-offs, names of personnel involved, test parameters, test results, pass/fail outcome, observations, faults or deficiencies, and any repairs or corrections made.

7.0 Document Control

7.1 **Approval:** GLOBAL_FAC_GFTT_APPR

7.2 **Notification**:
GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_EHS_TEAM_MEMBERS
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10W_FAC_NOTIFY
FCG_F10N_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MTTW_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
MTB_FAC_ALL_NOTIFY

7.3 **Final Viewer Access**
All Micron Team Members at all Sites

7.4 **Retention**
Adherence to the requirements specified in
Global Facilities - Document Control Procedures .

7.5 **Review**
Biennially (two years)

8.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document to replace previous document “PVC/CPVC Plastic Process Piping System Quality Control/Field Pressure Testing Procedure”, dated 07/29/16	05/15/2020

Appendix 1: Template Form for Site Specific Deviation List

Site:____________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT